

Predicting NBA Player-Game Field-Goal Attempts, Rebound Chances, and Rebounds with Similar-Player Matchups and Residual Learning

Malachi Fordham

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Abstract

This study develops and evaluates a pregame model for NBA player field-goal attempts (FGA), rebound chances, and rebounds. The approach separates opportunity from conversion. An interpretable first-stage model estimates a player’s recent stable workload, identifies statistically similar players, and uses the comparable players’ per-minute results against the same opponent to estimate an opponent effect. After sample-size shrinkage and bounding, that effect is applied to the target player’s recent per-game opportunity baseline. A second stage uses Elastic Net regression to learn residual errors left by this basketball baseline; rebounds are then obtained from corrected rebound chances and a separately learned rebound-conversion adjustment.

The study uses NBA regular-season player-game data from 2021–22 through 2025–26 and evaluates model choices with chronological, rolling outer-season tests. Across the 2023–24, 2024–25, and 2025–26 chronologically held-out test seasons, the selected opportunity model reduced equal-season mean absolute error by 2.36% for FGA, 4.22% for rebound chances, and 2.64% for rebounds relative to the tuned first-stage baseline. The learned conversion layer reduced rebound error by a further 0.86% across complete held-out seasons. A model trained only on normal-minute games modestly improved conditional FGA and rebound-chance accuracy. The final system selects Elastic Net over a competitive neural benchmark because it achieved nearly equal accuracy while remaining easier to interpret and substantially faster to fit.

1 Introduction

Single-game NBA box-score outcomes combine several mechanisms. A player’s field-goal attempts and rebound chances depend on workload and role, while actual rebounds also depend on how efficiently available chances are converted. Treating each result as one indivisible target obscures this structure.

For target player p , opponent o , and game date d , this project estimates

$$\widehat{FGA}_{p,o,d}, \quad \widehat{RC}_{p,o,d}, \quad \widehat{REB}_{p,o,d},$$

using only information available before tip-off. The design began as an interpretable comparison model: establish the target’s recent baseline, find players whose current workload and playing style are similar, measure how their opportunity changed against the target opponent, and transfer a carefully weighted portion of that effect to the target. Machine learning was added only after this rule-based model was complete, allowing the learned component to correct systematic errors without discarding the underlying basketball interpretation.

The study makes four practical contributions:

1. a reproducible definition of player similarity based on workload, opportunity, involvement, listed position, and starting role
2. an interpretable opponent adjustment with explicit sample-size trust and robust bounds
3. a leakage-controlled residual-learning framework evaluated on future seasons
4. separate all-games and normal-minutes final fitted models, with a learned rebound-conversion component

2 Research Questions and Hypotheses

The study examines five hypotheses. H3–H5 are evaluated through direct chronological model comparisons. H1 and H2 are design hypotheses supported by component behavior, but their isolated effects were not measured in the final outer-season experiment.

H1.

A player’s recent performance in games with a stable workload provides a useful baseline for predicting FGA, rebound chances, and rebounds.

H2.

Opponent-specific per-minute rates from statistically similar players improve the target baseline when the evidence is adjusted for sample size.

H3.

Machine-learning models can improve accuracy by learning systematic residual errors left by the baseline.

H4.

Training on games in which players received approximately their expected minutes improves predictions conditional on the player receiving a normal workload.

H5.

Modeling rebound conversion separately from rebound opportunity improves rebound predictions over using only the player’s recent conversion rate.

For rebounds,

$$\widehat{REB} = \widehat{RC} \times \widehat{ConversionRate}.$$

Per-minute rates are used to compare players with different workloads and to estimate opponent effects among similar players. Final V1 opportunity predictions apply those effects to the target player’s mean qualifying per-game FGA and rebound-chance baselines. Mean qualifying minutes are retained as a workload estimate and model feature, but V1 does not explicitly rescale the final prediction by projected minutes.

Listed position was initially considered as a hard comparison rule. Manual review showed that basketball role does not always follow position labels, so the final method uses position as a soft penalty within a broader statistical profile. This permits cross-position comparisons when workload, usage, touches, shooting volume, and rebounding opportunity are otherwise similar.

3 Data

3.1 Sources and unit of analysis

Data were obtained from NBA Stats through the community-maintained `nba_api` Python package [1]. A canonical warehouse joins regular-season base box scores, advanced usage statistics, and

player tracking statistics. Each row represents one player appearance in one game and includes the game date, teams, minutes, FGA, rebounds, rebound chances, touches, usage, listed position, and starter status.

3.2 Historical coverage

The local warehouse covers five complete regular seasons and 131,279 player-game rows.

Table 1: Canonical historical warehouse.

Season	Games	Players	Player-games	Tracking/usage coverage
2021–22	1,230	605	26,036	100%
2022–23	1,230	539	25,892	100%
2023–24	1,230	572	26,399	100%
2024–25	1,230	569	26,303	100%
2025–26	1,230	582	26,649	100%

The finalized 250-player all-games modeling dataset contains 59,156 eligible target player-games across 5,788 distinct NBA games. Its dates run from October 29, 2021 through April 12, 2026. The first days of a season are often ineligible because the player has not yet accumulated the required pregame history.

3.3 Data quality and leakage controls

The data pipeline validates required columns, nonnegative statistics, one row per (`game_id`, `player_id`), valid dates, and the expected season for each partition.

For every modeled game, the feature cutoff is the previous calendar day. Rolling baselines, player profiles, opponent rates, hyperparameter selection, and percentile bounds are all constructed from earlier data. Actual outcomes are attached only after pregame features are complete. The normal-minutes label uses actual game minutes solely to define research cohorts; it is never an inference feature.

4 Methodology

4.1 Recent target baseline

For a target game on date d , the model uses the 30 days ending at $d - 1$. Let M_i be minutes in each recent appearance. The initial workload estimate and qualification threshold are

$$M_{usual} = \text{mean}(M_i), \quad M_{min} = 0.75M_{usual}.$$

Games are retained when minutes are at least M_{min} and FGA and rebounds are observed. At least five qualifying box-score games and five qualifying games with rebound-chance tracking are required.

The baseline records mean qualifying per-game FGA and rebound chances, corresponding per-minute rates, mean qualifying minutes, and recent rebound conversion. The conversion estimate uses paired totals:

$$Conversion_{recent} = \frac{\sum_i REB_i}{\sum_i RC_i}.$$

Using totals gives high-opportunity games more influence than an unweighted mean of game-level conversion percentages.

4.2 Player profiles and similarity

Each recent player profile contains minutes, FGA per minute, rebound chances per minute, usage rate, touches per minute, starter rate, starter flag, and listed position. Numeric features are standardized across eligible players:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}.$$

For target t and candidate i , weighted numeric distance is

$$D_{numeric}(i, t) = \sqrt{\frac{\sum_j w_j (z_{ij} - z_{tj})^2}{\sum_j w_j}}.$$

Broad guard, forward, and center tokens are derived from listed position. A candidate with no token overlap receives a default position penalty of 0.35:

$$D_{final} = D_{numeric} + PositionPenalty.$$

Lower distance indicates greater similarity. The target is excluded from the candidate pool, and the default workflow selects the ten closest eligible players. Candidates must have complete recent profiles and valid FGA/minute and rebound-chances/minute observations against the exact target opponent within the same lookback window. Default distance weights are reported in Appendix A.

4.3 Opponent evidence, trust, and robust bounds

For comparable player s , metric m , and opponent o , the matchup ratio is

$$R_{s,m,o} = \frac{rate_{s,m \text{ versus } o}}{rate_{s,m \text{ overall}}}.$$

Player ratios are combined using the number of qualifying opponent games as weights:

$$R_{raw,m,o} = \frac{\sum_s n_{s,m,o} R_{s,m,o}}{\sum_s n_{s,m,o}}.$$

FGA and rebound chances retain separate sample counts. Before trust is applied, the raw multiplier is clipped to a robust interval:

$$R_{bounded} = \min(1.50, \max(0.50, R_{raw})).$$

Sample trust follows a piecewise smoothstep curve. The final anchors are 0.45 trust at five samples, 0.65 at ten, and 0.90 at fifteen or more. Within each interval, interpolation uses

$$S(x) = x^2(3 - 2x), \quad 0 \leq x \leq 1.$$

The effective multiplier is a convex combination of observed evidence and the neutral prior:

$$R_{effective} = 1 + w(R_{bounded} - 1).$$

With little evidence the baseline is preserved; as evidence increases, the effective multiplier moves toward the observed bounded ratio.

4.4 Interpretable V1 projection

The first-stage model predicts

$$\widehat{FGA}_{V1} = FGA_{baseline} \times R_{effective, FGA},$$

$$\widehat{RC}_{V1} = RC_{baseline} \times R_{effective, RC},$$

$$\widehat{REB}_{V1} = \widehat{RC}_{V1} \times Conversion_{recent}.$$

This model is independently usable and remains the reference prediction that the machine-learning stage attempts to improve.

4.5 V2 residual learning

Rather than replace the interpretable model, V2 predicts its historical error:

$$Residual_m = Actual_m - \widehat{m}_{V1},$$

$$\widehat{m}_{V2} = \max(0, \widehat{m}_{V1} + \widehat{Residual}_m).$$

Separate models correct FGA and rebound-chance residuals. Features summarize baseline quality, target role, comparison-pool quality, matchup evidence, and the V1 projections. The final opportunity schema contains 38 numeric features and one categorical listed-position feature. The feature groups are summarized in Appendix A; the complete machine-readable schema is retained in the model manifest and project repository.

Elastic Net combines L_1 and L_2 regularization [2] and is implemented with scikit-learn [3]. Median imputation, numeric standardization, and one-hot position encoding are fitted within each training period.

4.6 Learned rebound conversion

After rebound-chance correction, a separate chance-weighted Elastic Net model predicts an adjustment to the target’s recent conversion rate:

$$Correction = Rate_{actual} - Rate_{recent}.$$

Training rows are weighted by actual rebound chances so that low-exposure games contribute less conversion information. Zero-chance rows are excluded from conversion fitting but remain in final rebound evaluation. The predicted rate is clipped to training-only percentile bounds before final rebounds are calculated:

$$\widehat{REB} = \widehat{RC}_{V2} \times \widehat{ConversionRate}.$$

Trust-dependent fields are omitted from conversion features so that changing the final trust curve does not silently redefine the conversion model.

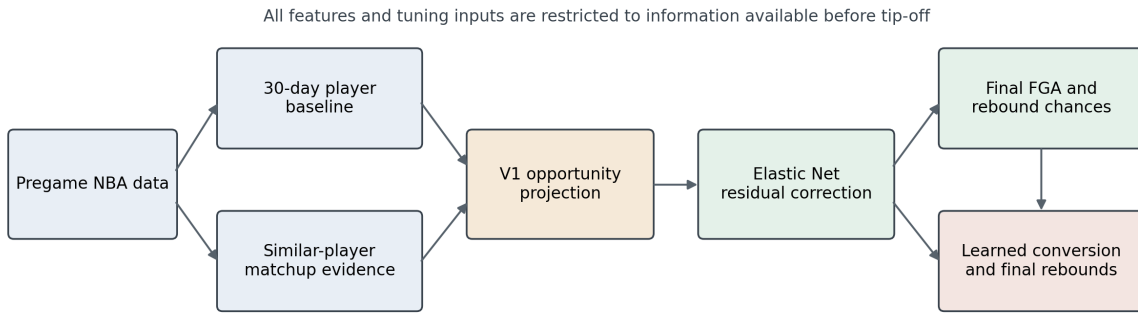


Figure 1: Final modeling pipeline. Opportunity and conversion are estimated separately, and every input is restricted to pregame information.

5 Experimental Design

5.1 Reproducible target-player pools

Convenience samples of well-known players would overrepresent particular roles. Eligible player-seasons therefore required at least 30 appearances, 20 average minutes, a 50% starter rate, at least 95% coverage for usage, tracking, and starter information, and known usage and broad position.

Players were stratified by broad position and low, medium, or high usage. Nested pools of 30, 60, 90, 150, and 250 target players were constructed with seed 42. The target pool determines which player-games become training rows; similar-player searches still use the full eligible league population available before each target game.

5.2 Compared model families

Three residual model families were evaluated:

- **Elastic Net:** regularized linear residual models with scaled numeric features and one-hot position;
- **Histogram Gradient Boosting:** nonlinear tree models using median imputation and absolute-error loss; and
- **Multi-task multilayer perceptron (MLP):** a shared 64-to-32 hidden representation with separate FGA and rebound-chance output heads, implemented in PyTorch [4]. Training used Huber loss, AdamW, early stopping, and a five-seed ensemble.

These comparisons tested whether model complexity added useful signal beyond the structured V1 features.

5.3 Chronological validation

Random row splits were avoided because later role information could then help predict earlier games. Development used walk-forward date splits. Final model comparison used three rolling outer seasons:

Table 2: Rolling outer-season evaluation folds.

Training seasons	Test season	Training rows	Test rows
2021–22 to 2022–23	2023–24	25,622	13,088
2021–22 to 2023–24	2024–25	38,710	11,099
2021–22 to 2024–25	2025–26	49,809	9,347

Within each fold, trust parameters and model parameters were selected using only the earlier training seasons, and the held-out season was used only for scoring that fold. Because the broader research design evolved through retrospective analysis of these experiments, the results should be interpreted as rolling out-of-time evaluation rather than preregistered untouched validation.

5.4 Metrics

Primary model selection used mean absolute error (MAE):

$$MAE = \frac{1}{n} \sum_i |\hat{y}_i - y_i|.$$

Supporting measures included root mean squared error, signed bias, equal-player MAE, seen/unseen-player subsets, 95th and 99th percentile absolute errors, maximum absolute error, and fitting/inference time. Conversion experiments additionally used player-clustered bootstrap confidence intervals. The headline values below are game-weighted metrics averaged equally across the three outer seasons.

Conversion uncertainty was estimated with 2,000 paired bootstrap replicates. Players were resampled as clusters within each test season, season-level MAE improvements were calculated, and the three seasons were averaged equally. Reported 95% intervals are percentile intervals generated with random seed 42.

6 Results

6.1 Sample scale and model selection

The initial five-player pilot contained only 182 complete rows and an imbalanced 34-row holdout; neither Elastic Net nor gradient boosting improved V1. A 90-player dataset expanded the sample to 18,021 rows, after which all three model families improved the baseline. This reversal established that the pilot was too small and role-imbalanced for residual learning.

On a common 9,347-row 2025–26 test set, increasing the Elastic Net target pool from 30 to 250 players generally improved MAE, with most accuracy gains reached by 60–150 players (Figure 2). The 250-player pool was selected because it provided the broadest tested training population with only modest additional fitting cost.

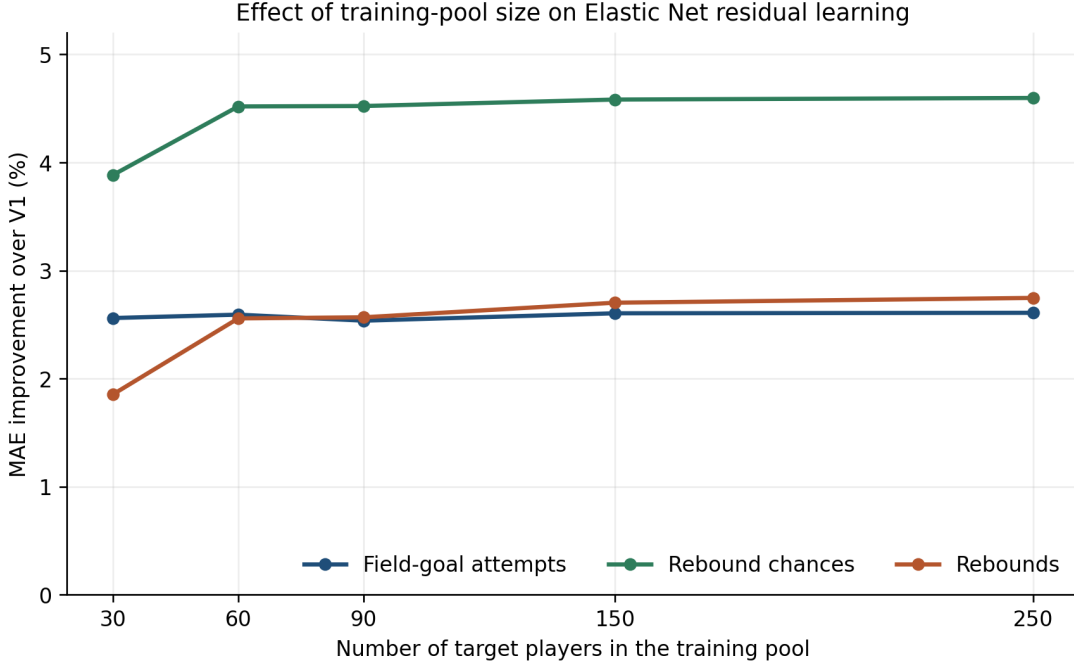


Figure 2: Learning curve for Elastic Net residual models on one common chronologically later 2025–26 test set. The curves show diminishing returns as the target-player training pool expands.

Across the rolling outer seasons, the Multi-task MLP was slightly better on FGA and rebounds but slightly worse on rebound chances. Its errors were about 99% correlated with Elastic Net errors, and it required substantially more training time. Elastic Net was therefore selected because the small average accuracy difference did not justify the neural model’s additional complexity and training cost.

Table 3: Equal-season model comparison across the three chronologically held-out test seasons.

Model	Output	Mean V1 MAE	Mean V2 MAE	Improvement
Elastic Net	FGA	3.136	3.062	2.36%
Elastic Net	Rebound chances	3.063	2.934	4.22%
Elastic Net	Rebounds	2.122	2.066	2.64%
Multi-task MLP	FGA	3.136	3.058	2.49%
Multi-task MLP	Rebound chances	3.063	2.936	4.13%
Multi-task MLP	Rebounds	2.122	2.065	2.70%

6.2 Chronological outer-season performance

The selected Elastic Net opportunity model improved all three outputs in each of the three chronologically held-out test seasons. Figure 3 shows the season-level results, while Table 3 reports equal-season means.

The largest repeatable gain occurred for rebound chances, the opportunity target most directly tied to the matchup and profile design. Rebound gains were smaller because opportunity error is compounded by conversion variance.

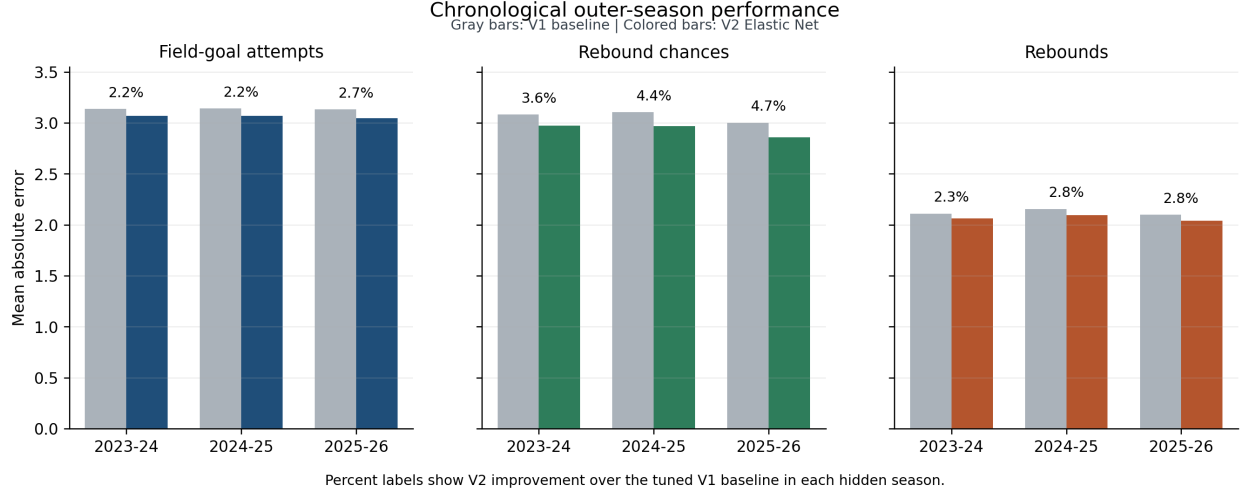


Figure 3: V1 and Elastic Net V2 game-weighted MAE in each chronological outer season. Axes begin at zero to avoid visually overstating the modest gains.

6.3 Normal-minutes experiment

A game was labeled normal when actual minutes were close to the pregame workload estimate. Here, $M_{expected}$ is the target player’s mean minutes across qualifying games in the recent baseline window. A game was considered normal when

$$|M_{actual} - M_{expected}| \leq \max(3, 0.15M_{expected}).$$

Approximately 59% of training and test rows met this condition. Evaluation was restricted to normal-minute games, while training used either all games or only normal-minute games.

Table 4: Conditional MAE on normal-minute games.

Training cohort	FGA MAE	Rebound-chance MAE	Rebound MAE
V1 reference	2.819	2.764	2.014
V2 trained on all games	2.786	2.691	1.987
V2 trained on normal-minute games	2.767	2.680	1.988

Normal-minute training modestly improved conditional FGA and rebound-chance accuracy, while the initial rebound result was essentially unchanged. The normal-minutes model was retained alongside the all-games model as a conditional projection. It estimates performance under the assumption that the player receives approximately the expected workload; it cannot determine before the game whether that condition will occur.

6.4 Robust multiplier bounds

The 0.50–1.50 interval affected fewer than 0.2% of complete-season rows. Average MAE and 99th-percentile absolute error changed only slightly, while the largest observed rebound-chance and rebound errors were substantially reduced. The bounds were therefore retained as protection against rare extreme matchup ratios rather than as a source of average-accuracy improvement.

Table 5: Effect of fixed 0.50–1.50 matchup bounds.

Diagnostic	Unbounded	Bounded
Mean FGA MAE	3.062	3.062
Mean rebound-chance MAE	2.934	2.931
Mean rebound MAE	2.066	2.065
Observed maximum rebound-chance absolute error	55.847	22.412
Observed maximum rebound absolute error	30.224	20.426
Mean FGA 99th-percentile absolute error	11.369	11.353
Mean rebound-chance 99th-percentile absolute error	11.187	11.156
Mean rebound 99th-percentile absolute error	7.916	7.899
FGA test rows clipped	0%	0.134%
Rebound-chance test rows clipped	0%	0.186%

The bounds are safety controls rather than filters. Raw ratios remain available for audit and as machine-learning features.

6.5 Rebound-conversion experiment

The recent player conversion rate was compared with position-prior shrinkage and a chance-weighted Elastic Net correction while holding the corrected rebound-chance forecast fixed.

Table 6: Equal-season rebound MAE by conversion method.

Method	All games	Gain	Normal minutes	Gain
Recent player rate	2.0645	–	1.9876	–
Position-prior shrinkage	2.0506	0.68%	1.9662	1.08%
Elastic Net adjustment	2.0467	0.86%	1.9631	1.23%

All three chronologically held-out test seasons improved under the learned adjustment. For the all-games setting, the mean reduction of 0.0179 rebounds had a player-clustered 95% bootstrap interval of approximately 0.0132 to 0.0225. In the normal-minutes setting, the mean reduction of 0.0245 had an interval of approximately 0.0189 to 0.0303. The effect is small in magnitude but repeatable.

7 Discussion

The strongest result is not a dramatic reduction in single-game error. It is the consistency of the residual-model improvements across chronological test seasons. Recent workload provides an interpretable starting point; per-minute rates support player comparison; statistical similarity makes manual comparison repeatable; and sample trust and bounds limit weak or extreme opponent adjustments.

Table 7: Interpretation of the five research hypotheses.

Hypothesis	Evidence	Assessment
H1	Recent stable-workload features formed a strong reference model, but a separate season-wide comparison against every simpler averaging baseline was not the focus of the final experiment.	Provisionally supported
H2	Similar-player matchup evidence provided an auditable opponent adjustment, while sample trust and bounds limited weak or extreme adjustments. Its incremental average value over a completely neutral matchup was not isolated in the final outer folds.	Partially supported
H3	Elastic Net residual correction reduced MAE for all three outputs in all three chronologically held-out test seasons.	Supported
H4	Normal-minute training improved conditional FGA and rebound-chance MAE; the first rebound comparison was flat, while learned conversion later improved conditional rebounds.	Partially supported
H5	Learned conversion reduced rebound MAE in every chronologically held-out test season, with player-clustered bootstrap intervals above zero.	Supported

Typical errors remained near three FGA, three rebound chances, and two rebounds. Important causes include unexpected playing time, injuries, foul trouble, blowouts, teammate availability, role changes, opponent shot volume and location, competition for the same rebound, imperfect player similarity, and irreducible single-game variation. These errors should be understood as the remaining uncertainty of a point forecast, not simply as a failure to fit a more complex algorithm.

Elastic Net was selected as the final model because it retained the basketball structure and matched the neural benchmark closely. The result is easier to audit, faster to retrain, and less operationally demanding. This does not imply that neural networks are generally unsuitable for basketball data; it means that the tested tabular feature set did not contain enough additional nonlinear signal to justify the more complex model.

8 Limitations

1. **Minutes and availability.** Expected minutes are based on recent qualifying games. The model does not ingest timestamped injury reports, confirmed starters, restrictions, coach reports, or depth-chart changes.
2. **Lineups and teammates.** Active-roster, lineup, on/off, and teammate-absence features are not represented, although they can sharply change usage and rebound role.
3. **Opponent representation.** Opponent effects are inferred from comparable-player outcomes rather than explicit pace, shot-location mix, scheme, team rebounding, or expected game environment.
4. **Similarity assumptions.** Weighted Euclidean distance is interpretable but imposes a simple geometry. Its weights were informed by basketball reasoning rather than optimized end to end.
5. **Rebound-chance quality.** Total chances do not capture contested status, chance distance, offensive/defensive split, opponent shot profile, or teammate competition.
6. **Point forecasts.** The system does not provide calibrated prediction intervals or threshold probabilities.
7. **Scope.** Evidence is limited to regular-season tracking-era data and a reproducibly selected

250-player target pool. It does not establish playoff accuracy or future-season stability.

9 Conclusion and Future Work

This project demonstrates that an interpretable basketball model and a modest machine-learning correction can work together. The selected system reduced average error relative to its tuned V1 reference in every chronological outer season, reduced observed extreme errors through robust matchup bounds, and found a small repeatable signal in rebound conversion. The gains are meaningful but limited: the model improves a noisy single-game forecast rather than making it certain.

The highest-priority extension is a pregame minutes and availability model using timestamped injury status, confirmed starters, recent rotations, return-from-injury indicators, and blowout risk. Further work should add calibrated prediction intervals, lineup context, rebound-chance quality, chronologically tuned similarity weights, and ongoing drift monitoring.

References

- [1] *nba_api: An API Client Package to Access the APIs for NBA.com*. GitHub repository and endpoint documentation. https://github.com/swar/nba_api.
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- [3] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [4] A. Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, 2019.

A Final Model Parameters and Feature Groups

A.1 Similarity weights

Table 8: Default numeric player-similarity weights.

Feature	Weight
Minutes	1.00
FGA per minute	1.25
Rebound chances per minute	1.50
Usage rate	1.25
Touches per minute	1.25
Starter rate	0.75

The default position mismatch penalty is 0.35, the default comparison pool is ten players, and a stricter optional setting can require the same starter flag.

A.2 Trust and final model parameters

The selected trust anchors were 0.45 at five samples, 0.65 at ten samples, and 0.90 at fifteen or more samples. Smoothstep interpolation was used between anchors, and matchup multipliers were bounded to 0.50–1.50 before trust was applied. The selected final model parameters are:

Parameter	Value
FGA Elastic Net α	0.005
FGA l_1 ratio	1.00
Rebound-chance Elastic Net α	0.005
Rebound-chance l_1 ratio	0.25
Multiplier bounds	0.50–1.50
Random state	42
All-games conversion α	0.003
All-games conversion l_1 ratio	0.10
Normal-minutes conversion α	0.003
Normal-minutes conversion l_1 ratio	0.25

A.3 V2 feature groups

The opportunity residual models use 38 numeric features and listed position as one categorical feature. Numeric features are organized into five groups:

Feature group	Numeric features
Recent baseline and workload	11
Target role profile	8
Comparison-pool diagnostics	7
Opponent matchup evidence	8
V1 projections	4
Total	38

The complete machine-readable feature schema is recorded in the model manifest and project repository.

B Reproducibility

The accompanying repository contains the source code, complete feature schema, NBA Stats endpoint details, automated tests, model manifests, and commands used to rebuild the historical warehouse, experiments, figures, and final fitted models.